Exp 4: Implement packet sniffing using Raw Sockets in python.

Aim:

To implement a packet sniffer using Raw sockets in python to capture and analyze network packets.

Procedure:

1) Open a terminal with root privilages

2) Create a python script 'sniffer.py'.

3) Import socket, struct, textwrap, and binascii Modules.

4) Create a raw socket using socket.AF_PACKET (Linux).

5. Bind the socket to the network interface.

6. Unpack the Ethernet header using struct.unpack().

7. For IP packets (EtherType 0x0800), unpack the IP header.

8. Run the script and observe captured packets.

Output:

```
# packet_sniffer.py

import socket, struct, binascii
def mac_addr(raw):
    return ':'.join(f'{b:02x}' for b in raw)
s = socket.socket(socket.AF_PACKET, socket.
                  SOCK_RAW,)
while True:
    raw, addr = s.recvfrom(65535)
    eth = struct.unpack('! 6s6sH', raw[:14])
    print(f'src: {src} : {dst} (proto: {proto:04x})')
    if proto == 0x0800:
        print(f' IP: {sip} -> {dip} TTL:{ip[5]}')
```

Result:

The packet sniffer was successfully implement. Ethernet frames and IP packets were captured, and source / destination MAC and IP addresses along with protocol information were displayed.